

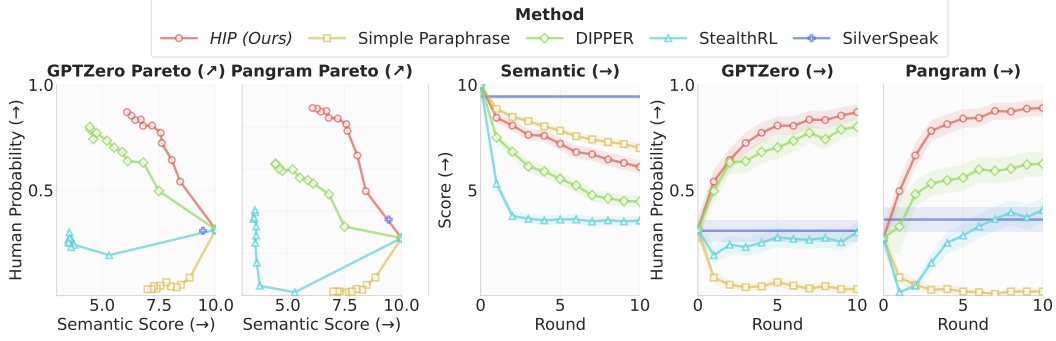


Figure 4: HIP versus baseline methods. The first two subplots show the GPTZero and Pangram Pareto frontiers for the trade-off between semantic preservation and human-likeness, while the last three show semantic score, GPTZero human probability, and Pangram human probability over the rounds, respectively. Shaded areas indicate 95% confidence intervals. HIP traces the strongest trade-off between semantic preservation and human-likeness among the methods we tested.

**Models.** We conduct experiments on both base and instruction-tuned models from the Qwen3 family [Yang et al., 2025] and the Llama3 family [Grattafiori et al., 2024]. For Qwen3, we use the 0.6B, 1.7B, 4B, 8B, and 14B models. For Llama3, we use the 8B and 70B models.

**Fine-tuning and inference configurations.** For each selected model, whether base or instruction-tuned, we apply the same minimal fine-tuning procedure on the training set  $\mathcal{D}$  to obtain a paraphraser  $\mathcal{M}_{\text{para}}$ , using the plain source-target format from Section 3.2. All runs use one epoch of training, a maximum sequence length of 2048, effective batch size 16, learning rate  $5 \times 10^{-5}$  with cosine scheduling, and LoRA [Hu et al., 2022] with rank 128, scaling factor 128, and dropout 0.05. For the 70B models, training uses QLoRA [Dettmers et al., 2023] for memory efficiency. Inference for all models is served with vLLM [Kwon et al., 2023]. At inference time, we apply  $\mathcal{M}_{\text{para}}$  iteratively for  $N = 10$  rounds. Across all runs, generation uses temperature 1.0 and top- $p$  0.95.

**Baseline methods.** We compare HIP against several representative detector-evasion baselines. The set of possible baselines is large and growing, and we do not aim to exhaust it. Instead, we choose the set of baselines that span different types of approaches and have released checkpoints:

- *Simple Paraphrase*: Directly applying a zero-shot paraphrase prompt at inference time.
- *DIPPER* [Krishna et al., 2023]: A supervised paraphrasing method that aims to preserve meaning while varying surface form, using lexical and sentence-level controls to steer diversity.
- *SilverSpeak* [Creo and Pudasaini, 2025]: A Unicode homoglyph-substitution method that perturbs token appearance without rewriting the text, targeting detector sensitivity to character-level cues.
- *StealthRL* [Ranganath and Ramesh, 2026]: A reinforcement-learning-based detector evasion method that optimizes a paraphraser against open-source detectors.

**Continuation evaluation.** For the continuation evaluation introduced in Section 1, we use 256 human-written and 256 AI-generated passages from the same selected RAID and MAGE domains. The prefixes are truncated to their first sentence and then used as continuation prompts. For each prefix, we generate one continuation and score only the generated text for human-likeness with GPTZero and Pangram. This evaluation compares the human-likeness of continuations from base and instruction-tuned models from the Qwen3 and Llama3 families as shown in Fig. 1. In Section A.1, we extend this setup to include HIP-adapted and continued-pretraining controls.

**Computation and API cost.** The experiments were conducted on GPU nodes with either 4 or 8 NVIDIA L40S GPUs. In total, the local training and inference runs consume roughly 300 GPU-hours. Across the project, OpenAI API usage for dataset construction, semantic scoring, and model fine-tuning cost roughly 200 dollars. At list prices, commercial-detector evaluation would have been

| Original AI Text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | MAGE / XSum | HIP (Llama3-8B) Round 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | MAGE / XSum |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <p><b>Semantic:</b> 10 (ref.)   <b>GPTZero:</b> 0.0%   <b>Pangram:</b> 0.0%</p> <p>A police force has apologised after its officers made a sandcastle “crime scene” featuring a naked dead woman. The incident happened during a training exercise on Bournemouth beach in Dorset, England. The force said the scene was “clearly inappropriate” and an investigation has been launched. It is not yet clear how the scene came to be created or who was responsible. The incident has caused outrage on social media, with many people calling for the officers involved to be disciplined. This is not the first time that the police have been accused of being insensitive in their training exercises. In 2016, the Metropolitan Police in London was criticised for using photos of real victims of knife crime in a training session.</p>               |             | <p><b>Semantic:</b> 8   <b>GPTZero:</b> 8.2%   <b>Pangram:</b> 0.0%</p> <p>A police force has apologised after its officers created a “crime scene” sandcastle, including a naked dead woman. Dorset Police said officers from its rural crime unit were taking part in a training exercise on Bournemouth beach in the south west of England. The force said the scene was “clearly inappropriate” and an investigation was under way. It is not yet known how the sandcastle was created or who was responsible. The incident has been condemned on social media and people have called for the officers involved to be disciplined. It is not the first time police have been accused of being insensitive in their training. In 2016, the Metropolitan Police in London faced criticism for using photos of actual victims of knife crime in a training exercise.</p>                                                        |             |
| HIP (Llama3-8B) Round 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | MAGE / XSum | HIP (Llama3-8B) Round 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | MAGE / XSum |
| <p><b>Semantic:</b> 9   <b>GPTZero:</b> 20.9%   <b>Pangram:</b> 100.0%</p> <p>A police force has apologised for the actions of officers who created a “crime scene” sandcastle including a naked dead woman. Dorset Police said officers from its rural crime unit were taking part in a training exercise on Bournemouth beach in the south west of England. The force said it was an “inappropriate” scene and was investigating. It is not known how the sandcastle was made and who was responsible. The incident has been criticised on social media, with people calling for the officers involved to be disciplined. It is not the first time police have been criticised for appearing insensitive in their training. In 2016, the Metropolitan Police in London was criticised for using photographs of victims of knife crime in their training.</p> |             | <p><b>Semantic:</b> 9   <b>GPTZero:</b> 100.0%   <b>Pangram:</b> 100.0%</p> <p>Dorset Police has apologised after officers were filmed building a sandcastle. It included the words crime scene and a naked dead woman. Dorset Police said it was “appalled” officers were caught on camera building the sandcastle on a rural crime team training day. The officers were at Bournemouth beach, in the south of England. Dorset Police said it was investigating the “inappropriate scene.” It was not clear how the sandcastle was constructed or how many officers were involved. Officers were criticised on social media, with many saying those involved should be disciplined. Police have been accused of being insensitive in the way they treat victims of crime and their families. The Met police in London was criticised heavily in November 2018 for using images of victims of knife crime in their training.</p> |             |

Figure 5: One qualitative Llama3-8B HIP trajectory from the main evaluation set, using an AI-generated source text from the xsum subset of the MAGE dataset. Each panel shows the text at a different round together with its semantic score and detector-assigned human probability from GPTZero and Pangram. This qualitative example illustrates how HIP can preserve the core content while progressively shifting detector judgments toward human.

more expensive: our GPTZero usage totaled about 40 million words, costing about 6000 dollars, and our Pangram usage totaled about 100000 passages, costing about 5000 dollars. GPTZero and Pangram provided research access to their models. To our knowledge, relatively few papers report detector-evasion results on state-of-the-art commercial detectors rather than only on open-source detectors, which strengthens the empirical relevance of our evaluation.

## 4.2 Experimental Results

**HIP humanizes AI-generated text across model families and scales.** We show in Fig. 3 the results of applying HIP to base and instruct checkpoints from the Qwen3 and Llama3 families. Each subplot represents one model family, and each line represents one model size. Within each subplot, the first two panels show the GPTZero and Pangram Pareto frontiers for the trade-off between semantic

preservation and detector evasion, while the last three show how semantic score and detector-specific human probabilities change over iterative rounds of paraphrasing.

The main pattern is consistent across all four families. After training with HIP, as the paraphraser is applied for more rounds, detector-assigned human probability rises on both GPTZero and Pangram, while semantic fidelity gradually declines. In other words, the method works by moving model outputs toward a more human-like region of the detector space, but it does so at a semantic cost. This trend holds for both base and instruction-tuned checkpoints, which indicates that the humanization effect of HIP is not specific to a single model family, size, or post-training state.

Model size primarily affects the trade-off when the size is low, rather than improving it uniformly. Within Qwen3, moving from the smaller checkpoints to 4B materially improves the trade-off curve, but beyond 4B the frontiers shift only modestly and not necessarily for the better. Llama3 shows the same qualitative pattern: the 70B models are slightly more semantically stable than the 8B models, but both achieve similar trade-offs. Our interpretation is that HIP mainly requires a model large enough to paraphrase competently. Once that threshold is reached, the method largely works.

**Qualitative examples.** Aggregate trade-off curves correspond to recognizable local edits at the example level. Figure 5 shows one Llama3-8B HIP trajectory from the main evaluation set. Across rounds, the model preserves the core factual content while progressively rewriting phrasing and local structure. In this example, both detectors initially flag the source text as AI-generated. By round 2, GPTZero has moved into a more ambiguous regime and Pangram has already flipped to human. By round 10, semantic fidelity remains high while both detectors score the output as human. In Section B, we present additional qualitative examples from HIP as well as the baseline methods.

**Baseline comparison.** We compare HIP against all baselines in Fig. 4. For every model-based baseline, we evaluate the released checkpoint from the original paper, and Simple Paraphrase is the only prompting-based baseline. As shown in Fig. 4, under the commercial detectors we study, only DIPPER still achieves a non-trivial semantic-evasion trade-off among the methods we tested. SilverSpeak raises Pangram human probability slightly, but has little effect on GPTZero. Both Simple Paraphrase and StealthRL fail to achieve a meaningful trade-off on either detector. We therefore use Llama3-8B as HIP’s comparison point, since it is smaller than DIPPER’s 11B backbone and thus provides a conservative comparison. As Fig. 4 shows, HIP on Llama3-8B substantially outperforms all tested baselines, achieving the strongest trade-off frontier under both detectors.

**Ablation studies and additional experiments.** We defer additional experiments to Section A, including details about continuation evaluation (Section A.1), HIP on OpenAI models via the fine-tuning API (Section A.2), running HIP on instruct models with native chat templates (Section A.3), and output-layer-only adaptation (Section A.4).

## 5 Discussion

**Why HIP works.** Our results suggest that detector-judged human-likeness improves when the model is trained directly on human tokens. Base models have this property because pretraining is almost entirely human text. HIP appears to recover part of the same behavior, but now under a paraphrase objective. The model is updated on paired rewrites whose targets are human-written passages, and iterative application reinforces that effect by progressively removing AI-generated context from the passage. On this view, HIP works not because it discovers a special decoding trick, but because it reintroduces a human-text training signal while enabling iterative paraphrasing.

**Implications for detectors.** Our findings suggest that current detectors are not primarily detecting “AI” in general. Rather, they appear to be sensitive to the statistical effects of post-training and to the context in which a passage is generated. This motivates two research questions for detector design:

1. Does post-training inherently leave detector-visible statistical fingerprints, or are current alignment pipelines simply introducing artifacts that better training procedures could avoid?
2. Can current statistical or neural detector paradigms reliably identify text from true base models, rather than primarily identifying post-training distortions?